Introduction to Biomedical Engineering

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Introduction to Biomedical Engineering (A66666)

Short Title:	Intro to Bio-Med		
Faculty	Engineering		
Department	Engineering Technology		
Cluster	Engineering		
Author			
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module aims to advance the students' knowledge of the fundamentals of biomedical engineering; specifically in respect of the analysis and evaluation of complex biomedical devices. A comprehensive review of the design, operation and manufacture of advanced biomedical engineering technologies (electromechanical domains) is undertaken; with particular emphasis on associated industry driven technology, business and regulatory roadmaps. \emph{Bachelor of Science}\emph{(Honours)}\emph{in Physics for Modern Technology} - Stage 4 Semester 8 (WD_KPHTE_B) [Elective] Biomedical Engineering -- Past, Present and Future Advanced Biomedical Instrumentation [electro-mechanical domains] Biomechanical Systems & Technologies [materials, cellular/tissue/organ mechanics and associated technologies] Cardiovascular Systems & Technologies [materials, cellular/tissue/organ mechanics and associated technologies] Advanced Biotechnologies [drug delivery, tissue engineering, nanobiotechnology] Demonstrate a knowledge and comprehension of the fundamental concepts of biomedical engineering as applied in the analysis of advanced biomedical devices. Demonstrate an ability to apply the knowledge and comprehension gained in analysing and relating macroscopic biomedical engineering and device performance parameters to fundamental biomedical engineering theory and the associated manufacturing processes. Demonstrate an ability to evaluate partial and/or full design criteria as used in the development, manufacture and use of advanced biomedical devices. Various Journal & Conference Publications (directed).

Programmes

	BEng (Hons) in Mechanical & Manufacturing Engineering (`WD_EMSEN_B`)	4 / 8 / E
`ENGR-0157`	BSc (Hons) in Physics for Modern Technology (`WD_KPHTE_B`)	4 / 8 / E

Indicative Content

Learning Outcomes

On successful completion of this module, a student will be able to:

Learning and Teaching Methods

- Lectures
- Case studies

  Practical's/Project Work

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	50%	
Case studies/Practical/Project work	50%	1,2,3
Final Written Examination	50%	1,2

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

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Enderle, J., S. Blanchard and J. Bronzino. Introduction to Biomedical Engineering. .: Academic Press, 2005.

Saltzman, W.M. Biomedical Engineering. Cambridge: Cambridge University Press, 2009. Schatzlein, A.G. and I.F. Uchegbu. Polymers in Drug Delivery. .: CRC Press, 2006.

Smallwood, D.C. Barber, P.V. Lawford and D.R. Hose. Medical Physics and Biomedical Engineering. .: Taylor Francis, 1999.

Adam, A. Textbook of Metallic Stents. .: ISIS Medical Media Ltd., 1997.

Akmal, N., W.T. Law and A.M. Usmani. Biomedical Diagnostic Science and Technology. .: Informa Healthcare, 2002.

Requested Resources

- Lecture Room: Loose Seated